# Whitmore Capital Partners SAA/TAA Mandate

## FIN 496 Foundation Project

- External consultant recommendation for a fictional $1.8B single-family office.
- Pipeline is fully reproducible from `python taa_project/main.py`.
- Final run mode: --timesfm.

# Mandate & IPS Highlights

- Target 8% return with 15% volatility ceiling and 25% max drawdown tolerance.
- Core must stay at or above 40%, Satellite at or below 45%, Non-Traditional at or below 20%.
- No shorts, no hard-coded risk-off safe haven, and 5 bps round-trip cost applied at each rebalance.

# SAA Method Choice

- Chosen SAA method: constrained risk parity.
- Reason: it clears the 8% return mandate while staying below the 15% volatility ceiling without leaning on unstable mean forecasts.
- Alternatives were evaluated side-by-side in the diagnostics notebook and `saa_method_comparison.csv`.

# TAA Signal Architecture

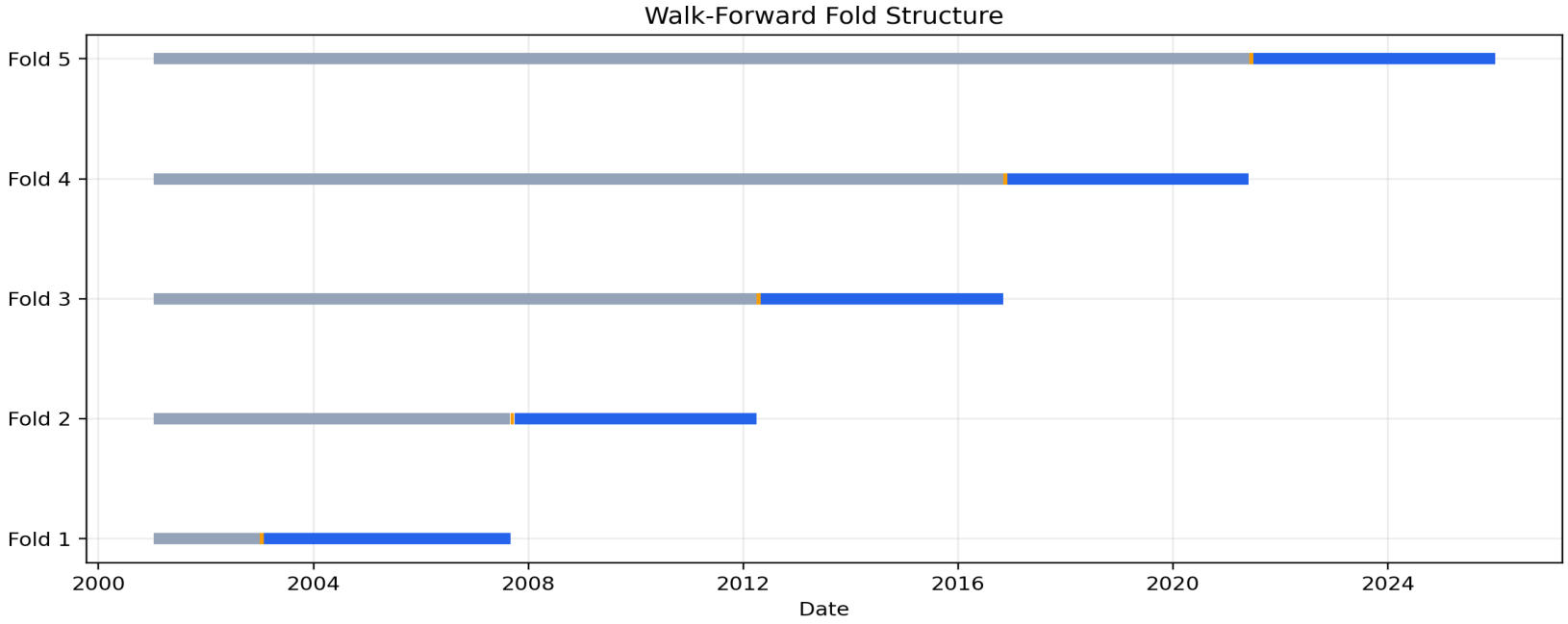
- Lagged macro -> 3-state HMM regime probabilities.
- Daily price histories -> Faber trend + ADM momentum.
- Optional TimesFM forecast -> direction and volatility input.
- Signal ensemble -> cvxpy optimizer inside TAA bands with turnover and volatility controls.

# Risk Overlays & Vol-Budget Sweep

Config comparison figure not available for this output directory.

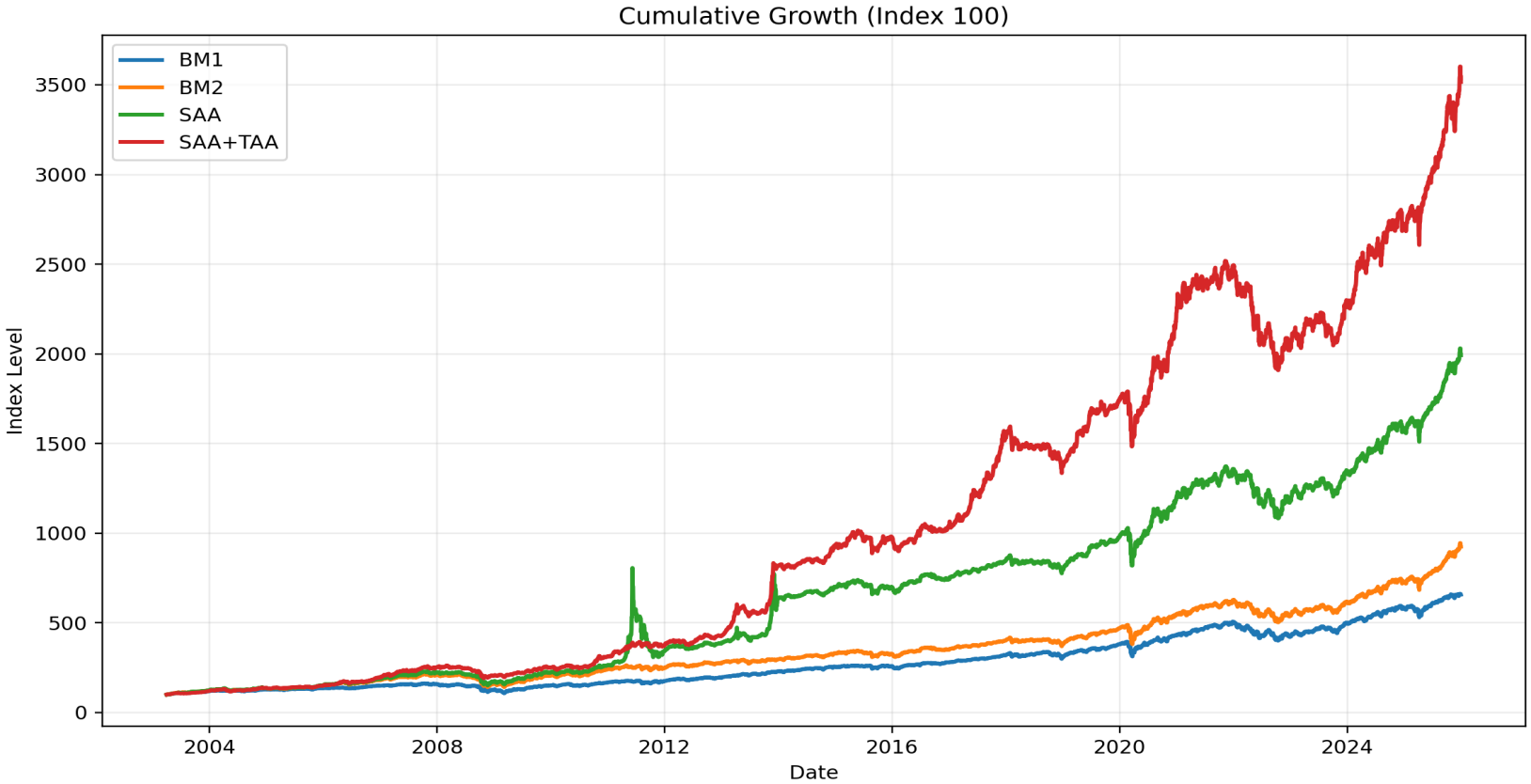
- Six canonical configurations are compared here once the sweep artifacts are present.
- Overlays only tighten the risk envelope; they do not hard-code a safe-haven allocation.

# Walk-Forward Design




Deflated Sharpe Ratio: 1.000 across 127 disclosed TAA trials.

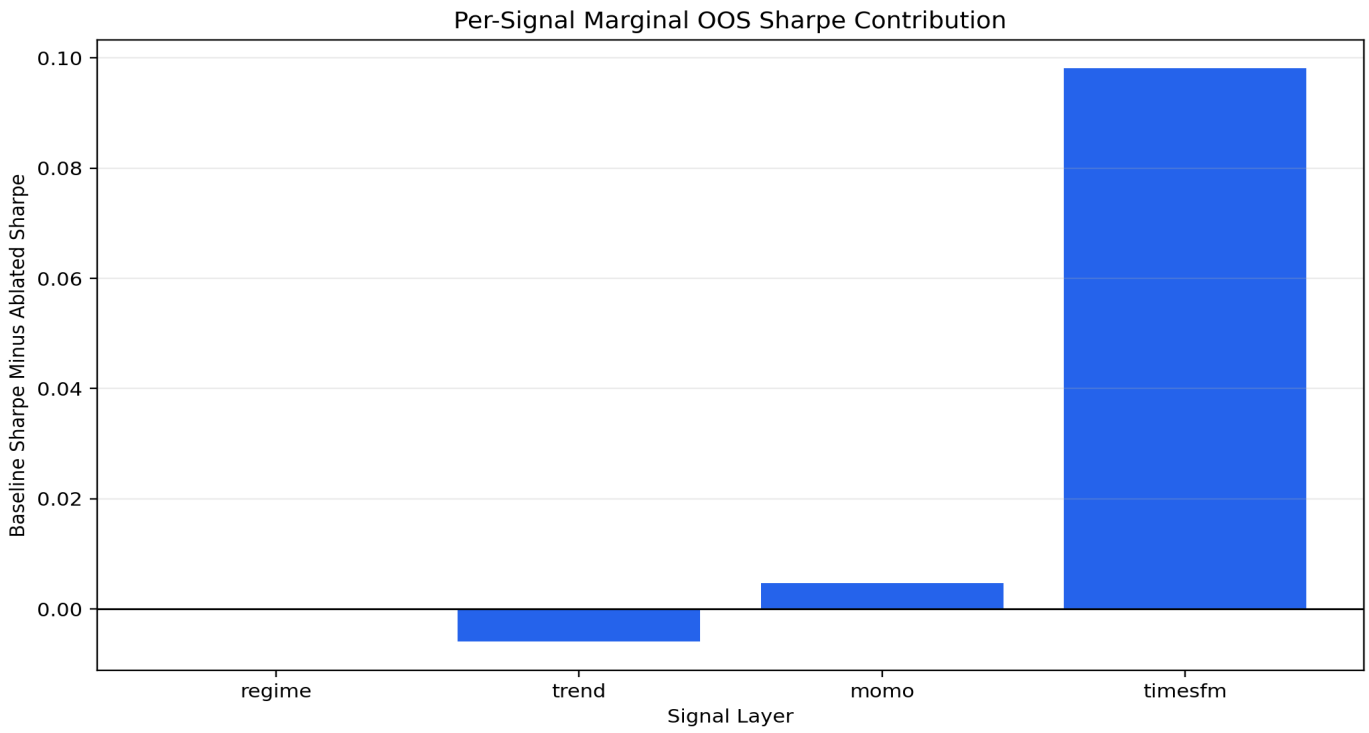
# Cumulative Growth



# Metrics Table

| portfolio | annualized_return | annualized_volatility | Sharpe | Sortino | Calmar | max_drawdown |
|---|---|---|---|---|---|---|
| BM1 | 0.054 | 0.096 | 0.349 | 0.489 | 0.158 | -0.339 |
| BM2 | 0.079 | 0.096 | 0.611 | 0.854 | 0.223 | -0.352 |
| SAA | 0.088 | 0.144 | 0.474 | 0.715 | 0.143 | -0.618 |
| SAA+TAA | 0.138 | 0.098 | 1.206 | 1.756 | 0.502 | -0.275 |

# Attribution




- Separate decomposition delivered for SAA vs BM2 and TAA vs SAA/BM1/BM2.
- Signal value is measured with leave-one-out OOS reruns, not static coefficient inspection.

# Limitations

- HMM states are statistical and can drift.
- TimesFM is optional and not finance-native.
- Covariance shrinkage and optimizer penalties remain engineering choices, not immutable truths.

# Recommendation

- SAA+TAA annualized return: 13.79% versus 8.83% for SAA and 7.86% for BM2.
- Use the risk-parity SAA as the policy anchor and keep the TAA layer conditional on continuing OOS superiority after costs and DSR adjustment.
- Track turnover and stressed-regime attribution before expanding risk budget to the overlay.

# Q&A;

- Key files: `taa_project/main.py`, `taa_project/analysis/reporting.py`, `taa_project/report/build_report.py`.
- Key artifacts: report PDF, deck PDF, diagnostics notebook, trial ledger, IPS audit.